

Pre-RGY Conversation to RGY Capsule

This is the scoped Phase 1 architecture. It covers the landing conversation, anonymous and signed-in memory, consent, perception, user understanding, safety, classification, and creation of an RGY capsule when a real signal emerges. It stops before execution workspaces, publishing, workflow automation, and commercial systems.

Phase 1 only

No autonomous execution

No execution workspace

No commercial build

Capsule-ready boundary

1

Before a goal exists

CubiQo is conversational: it listens, responds, remembers safely, and learns the user.

2

When a signal appears

Understanding/classification decides whether this is chat, memory, journal context, or an RGY activity signal.

3

When strong enough

The system writes a structured RGY capsule: color, keyword, domain hint, confidence, status, and user-visible summary.

4

Boundary

Phase 2 will decide what to do with the capsule. Phase 1 only prepares it correctly.

CANONICAL PHASE 1 PIPELINE

The pipeline is intentionally not "execution first." It is understanding first. A goal or capsule is an outcome of the conversation, not a requirement for every message.

Everything before execution

Input

Text, voice, image, camera frame, browser/screen context when user provides it.

Permission

Mic, camera, memory, location, push and tracking consent are checked before use.

Perception

Speech-to-text, vision, OCR-like visible text, screenshot understanding.

Context

Session, time, language, anonymous memory, user profile, journal, signed-in memory.

Understand

Intent, emotional tone, topic, goal hint, user preference, risk, and freshness need.

Safety

Input safety, age-gated/hard-block/crisis checks, output safety for sensitive paths.

Classify

Conversation, memory update, journal insight, onboarding fact, or RGY signal.

Capsule

If useful: save RGY capsule with confidence and status. Phase 1 ends here.

VISUAL FLOW

Three user states in one Phase 1 map

This is the relaxed board version: first-time anonymous user, returning anonymous user, and signed-in engaged customer all flow into the same understanding and capsule boundary.

USER INTERACTION AT LANDING APP

**FIRST-TIME
ANONYMOUS
USER**

NO SERVER MEMORY START LOCAL SESSION ONLY

localStorage session_id, first_seen_at, visit_count, language hint. No DB write.

**RETURNING
ANONYMOUS
WITHIN TTL**

CUBIQO REMEMBERS LIGHTLY LIKE A FRIEND, NOT A DATABASE

Use local signals and summary only. Nudge to sign up before memory expires.

**SIGNED-IN
RETURNING
ENGAGED USER**

DURABLE MEMORY + JOURNAL PROFILE IS UPDATED OVER TIME

Supabase reads profile, memory_events, user_ai_profile, journal insights, signals.

**DB / ENV
LOCAL FILES
RUNTIME CONFIG**

**APIS
AUTH
VOICE
VISION**

**UNDERSTANDING + CLASSIFICATION
RGY IS ONE POSSIBLE OUTPUT**

DECIDE: CHAT, MEMORY, JOURNAL, ONBOARDING, OR CAPSULE

**NORMAL
CONVERSATION**

**MEMORY
UPDATE
PROFILE HINT**

**RGY SIGNAL
IF STRONG
ENOUGH**

**CAPSULE
READY + SAVED
PHASE 1 ENDS**

OUTPUT: useful response now, not execution

Guardrails: no payments, no final send/submit/push

What happens for each user type

1. Anonymous first-timer

No account, no durable profile, no server memory. CubiQo starts with a clean conversational opener and creates a temporary local session.

Storage

`localStorage.cubiqa_session` only.

Memory

None at first. Light summary can form inside the browser session.

Goal

Not assumed. CubiQo detects, asks, and learns gradually.

Capsule

Can create a local capsule preview, but durable save requires sign-in.

2. Anonymous returning user

The user comes back before local memory expires. CubiQo can reference the last topic lightly and invite account creation when value is clear.

Storage

Local TTL memory, recent signals, optional last summary.

Greeting

Specific but not creepy: "Still thinking about the store idea?"

Conversion

"Want me to keep this longer?" only after enough value exists.

Migration

On sign-up, local memory migrates into Supabase tables.

3. Signed-in engaged user

Durable memory, profile, journal insights, and prior signals become part of the context bundle before CubiQo answers.

Storage Supabase: profile, signals, memory events, user AI profile, journal.

Greeting Text-first greeting; voice only after audio unlock.

Understanding Language, time, location if permitted, user style, recurring blockers.

Capsule Strong activity signals become durable RGY capsules.

What sits where

Frontend

- Landing chat surface.
- Anonymous local session manager.
- Voice input capture and audio unlock gate.
- Image/camera upload frame capture after permission.
- Right-side RGY preview panel only when a signal exists.
- Sign-up migration trigger for local memory.

API Routes

- `/api/agent/stream` - conversational answer with context bundle.
- `/api/agent/greet` - text greeting; voice after browser unlock.
- `/api/agent/onboard` - light profile bootstrap for signed-in users.
- `/api/rgy/classify` - safety plus understanding plus RGY signal detection.
- `/api/perception/vision` - structured visual context from image/screenshot.
- `/api/memory/session-close` - writes summaries and commitments.

Supabase Tables

- `profiles` - display name, language, timezone, consent hints, onboarding state.
- `signals` - RGY activity signals and capsule source records.
- `memory_events` - session summaries, commitments, blockers, outcomes, episodes.
- `user_ai_profile` - CubiQo's inferred understanding with confidence and corrections.
- `journal_entries` - user-authored journal plus extracted memory insights.
- `user_permissions` - mic, camera, memory, location, push statuses.
- `safety_patterns` and `system_prompts` - config-driven safety and behavior.

Config and Env

- `OPENAI_API_KEY` - conversation, structured understanding, vision, STT as configured.
- `ELEVENLABS_API_KEY` - optional TTS after user interaction unlock.
- `NEXT_PUBLIC_SUPABASE_URL` and keys - auth and DB access.

- `NEXT_PUBLIC_BASE_URL` - required; never silently fallback to production.
- `RGY_MODEL`, `OPENAI_MODEL`, `UTILITY_MODEL` - runtime model routing.
- `HISTORY_WINDOW_SIZE`, `MEMORY_EVENT_LIMIT` - tunable limits.

MEMORY MODEL

User memory before and after sign-up

This is the center of Phase 1. CubiQo needs a trustworthy understanding of the user before it can safely route goals in later phases.

Memory lane	Anonymous	Signed-in	Update rule
Session memory	localStorage with short TTL	Conversation/session state plus DB summary	Updated during chat; summarized at session boundary
RGY signals	Local preview only	<code>signals</code> rows	Written when a strong activity signal appears
Memory events	Not server-side until signup	<code>memory_events</code>	Commitments, blockers, outcomes, pivotal episodes, session summaries
User AI profile	Not durable	<code>user_ai_profile</code>	Bootstrapped on onboarding; refreshed after meaningful sessions
Journal	Not durable	<code>journal_entries</code>	User-authored; extracted into memory only with consent and care

UNDERSTANDING AND CLASSIFICATION

RGY is an output, not the whole brain

Conversation

Normal chat stays normal chat. CubiQo answers, asks clarifying questions, reflects preferences, and may update light memory.

No capsule required

Memory update

If the user reveals stable facts, preferences, commitments, or blockers, CubiQo records them with confidence and source.

Profile + memory

RGY capsule

If a meaningful activity signal emerges, CubiQo writes color, keyword, intent hint, safety state, confidence, and summary.

Capsule-ready

Classification contract

Safety always runs first. Then CubiQo decides whether the input is ordinary conversation, onboarding material, a memory event, a journal insight, a question needing fresh context, or a true RGY signal. Only the last path creates a capsule.

The two hard rules still apply in Phase 1

1. Do not touch payments

Phase 1 does not initiate purchases, billing, checkout, payment setup, or financial transactions. It can discuss concepts but does not act.

2. Do not final-submit anything

No autonomous send, submit, publish, push, deploy, or final external action. The user keeps the last button in later phases too.

Safety layers included here

Input safety, output safety for sensitive outputs, consent checks before sensors, RLS for user-owned rows, and no silent global connector fallback. These are Phase 1 foundations because trust starts before any external execution exists.

What to build now

Must build

- Anonymous local memory with TTL and sign-up migration.
- Signed-in onboarding conversation.
- `memory_events` and `user_ai_profile` tables with RLS.
- `user_permissions` table for mic, camera, memory, location, push.
- Text-first greeting and audio unlock before TTS playback.
- Reliable STT path and optional Web Speech fallback.
- Vision route returning structured visual context, not just a caption.
- Understanding/classification route with RGY as one possible result.
- Capsule save boundary: color, keyword, confidence, status, summary.

Explicitly not Phase 1

- Task execution and route planning.
- Structured execution workspace.
- Domain-specific execution flows.
- Browser automation for acting on external sites.
- Publishing, submitting, deploying, or final-button actions.
- Commercial systems, checkout, and revenue analytics.
- Market watch, background briefings, intervention systems.

MASTER CHECKLIST

Phase 1 done means this works

Area	Definition of done
Anonymous first visit	New local session created. No DB write. Neutral greeting. User can chat normally.
Anonymous return	Within TTL, CubiQo references prior topic lightly. After TTL, clean slate.
Sign-up migration	Local signals and summary migrate into Supabase. localStorage clears after successful write.
Onboarding	Signed-in user gets short conversational onboarding. Primary goal and initial user_ai_profile are written.
Memory	Session summaries, commitments, blockers, journal insights, and pivotal episodes are saved with source and confidence.
Permissions	Mic/camera/location/memory preferences are recorded and respected before each use.
Understanding	System separates ordinary chat, memory updates, journal insights, onboarding facts, and RGY signals.
RGY capsule	Strong signal creates durable capsule record with color, keyword, intent hint, confidence, and summary.
Safety	Input safety, output safety, RLS, no silent fallback, no payments, and no final submissions verified.

COVERAGE AUDIT

Does this cover the large plan you pasted?

Mostly yes for the pre-capsule and capsule foundation. No for execution, revenue, background automation, and operational dashboards. Those are intentionally not designed inside this Phase 1 master.

Area from the larger plan	Status in this master doc	Why
Input: voice, text, image, camera, browser context	Covered	Included as the Phase 1 perception layer, with permissions before sensors.
Perception layer: STT, vision, screenshot understanding	Covered	Included, with structured visual context as a target rather than plain captions.
Context layer: time, language, location, memory, user profile	Covered	Included as the context bundle for signed-in and anonymous states.
Daily context / fresh world state	Partial	Phase 1 stores the slots and context concept. Scheduled research/digest jobs are later.
RGY classification	Covered	Reframed correctly: RGY is one output of Understanding + Classification, not every message.
Signals and capsule creation	Covered	Phase 1 ends at a saved, structured capsule with color, keyword, status, summary, and confidence.
Memory events and AI-maintained user profile	Covered	Included as core Phase 1 tables and behavior.
Journal influence on user understanding	Covered	Included as a memory lane, with extracted insights feeding memory only with care.
Anonymous local memory and sign-up migration	Covered	Included for first-time, returning, and signed-in user flows.

Area from the larger plan	Status in this master doc	Why
Onboarding conversation	Covered	Included as a required signed-in bootstrap path, not a form.
Permissions and consent	Covered	Included as a first-class layer and Supabase table.
Input and output safety	Covered	Included as Phase 1 trust foundations.
Audio greeting and ElevenLabs	Partial	Text-first greeting and audio unlock are included. Voice generation implementation belongs in build tickets.
Rate limiting and API usage cost protection	Partial	Recognized as foundation, but detailed billing/subscription tiers are outside Phase 1.
Background briefings, interventions, market watch, cron refreshes	Later	These begin after a capsule exists. They are Phase 2+ execution/proactive systems.
Execution workspace / dashboard after capsule	Later	Phase 1 only prepares the capsule. The workspace that acts on it should be a separate document.
Commerce recommendations, cards, coupons, tracked links	Later	This is important, but not part of pre-capsule memory/classification. It depends on trustworthy memory first.
Direct partners, broad affiliate networks, product APIs, postbacks	Later	Commercial routing belongs after the recommendation quality gate exists.
Subscription checkout and paid tiers	Later	Commercial packaging is deliberately out of this Phase 1 master.
Admin monetization dashboard	Later	Operational metrics are necessary later, but not required to reach capsule-ready Phase 1.
GDPR export/delete, offline fallback, PWA/mobile hardening	Partial	The trust need is noted. Full product/legal implementation should become separate tickets.

SINGLE SOURCE

How to use this doc

This file is now the Phase 1 master visual and engineering scope: `docs/cubiqo-capability-engine-future-state.html`. Use it for conversations, edits, screenshots, and planning until Phase 1 is locked. Phase 2 should get a separate document later so the execution layer does not contaminate the memory/classification foundation.